Technology Stack (Architecture & Stack)

Date	25 September 2026	**Team ID**	SWTID-2026-4307
Project Name	AI Weather-Wise API	**Maximum Marks**	4 Marks

Technical Architecture & Guidelines

Guidelines for Architecture & Stack:

- Include all processes (Application logic / Technology Block).
- Provide infrastructural demarcation (Local / Cloud).
- Indicate external interfaces (Third-party APIs etc.).
- Indicate Data Storage components / services.
- Indicate interface to machine learning models (if applicable).

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with application (Web Dashboard, Mobile App, API Portal)	React.js / Angular, HTML5, CSS3, JavaScript
2	Application Logic - 1	Core Backend API gateway & routing logic	Python (FastAPI / Flask) / Node.js
3	Application Logic - 2	Weather data processing & Transformation Engine	Python (Pandas, NumPy, Scikit-learn)
4	Application Logic - 3	Automated Notification & Alert Dispatcher Service	Celery / Redis Task Queue
5	Database	Relational database for users, API keys, and operational data	PostgreSQL / MySQL
6	Cloud Database	Scalable Cloud Managed NoSQL/TimeSeries Database for historical weather logs	MongoDB Atlas / AWS DynamoDB / InfluxDB
7	File Storage	Storage for historical dataset backups and ML model artifacts	AWS S3 / Cloud Object Storage
8	External API - 1	Live Satellite & Global Weather API data feeds	OpenWeatherMap API / IBM Weather API
9	External API - 2	Geocoding & Location services API	Google Maps / OpenStreetMap API
10	Machine Learning Model	Microclimate weather forecasting & predictive risk model	LSTM / XGBoost Neural Networks (TensorFlow/PyTorch)
11	Infrastructure (Server / Cloud)	Containerization, deployment, and cloud infrastructure	Docker, Kubernetes, AWS / Cloud Foundry

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open Source Frameworks	Open-source frameworks used for rapid development and flexibility	FastAPI, React.js, Scikit-learn, TensorFlow

S.No	Characteristics	Description	Technology
2	Security Implementations	Access control, data encryption, and protection mechanisms	SHA-256 Encryption, JWT Authentication, OAuth2, OWASP Guidelines
3	Scalable Architecture	Microservices-based 3-tier scalable architecture design	Docker Containers, Kubernetes Auto-scaling
4	Availability	High availability through load balancing and failover mechanisms	NGINX Load Balancers, Multi-Region Cloud Deployment
5	Performance	Optimization for low-latency weather API responses	Redis Cache, Cloudflare CDN